Building a Zero Trust Network Environment with OpenZiti

Secure and safe networks are the need of the hour. This brief tutorial explains in detail how OpenZiti, an open source software-defined networking platform, can be used to set up a Zero Trust network environment.

As a security model, Zero Trust adheres to the tenet of "never trust, always verify" while granting access to any network resource. This ensures that the system will insist on authentication and authorisation before allowing any user or device, no matter how legitimate they may seem, to access sensitive data or network resources.

For building safe virtual networks, OpenZiti provides an open source software-defined networking (SDN) platform. Ziti networks are virtual networks that network managers may set up and maintain with ease. These networks can link devices and apps together, no matter where they are. Connecting and securing Internet of Things (IoT) devices, cloud services, and on-premises applications has never been easier than with OpenZiti. To protect information while it is in transit, it employs a Zero Trust security approach and offers end-to-end encryption.

Why OpenZiti?

OpenZiti's design aims to reduce the attack surface because it is based on zero trust principles. In what ways does it achieve that? By keeping all services and ports hidden from the public internet. Contrast this with virtual private networks (VPNs), which often have vulnerable open ports. This smaller assault surface is advantageous.

OpenZiti employs intelligent routing. Optimal latency and bandwidth are achieved by it choosing the optimum path for traffic. This is useful if network performance is paramount in dispersed systems or worldwide applications.

Additionally, it supports hybrid and multi-cloud environments. Having a system that can be easily configured to work across diverse environments is becoming more valuable as more and more organisations migrate their operations to the cloud. With OpenZiti,

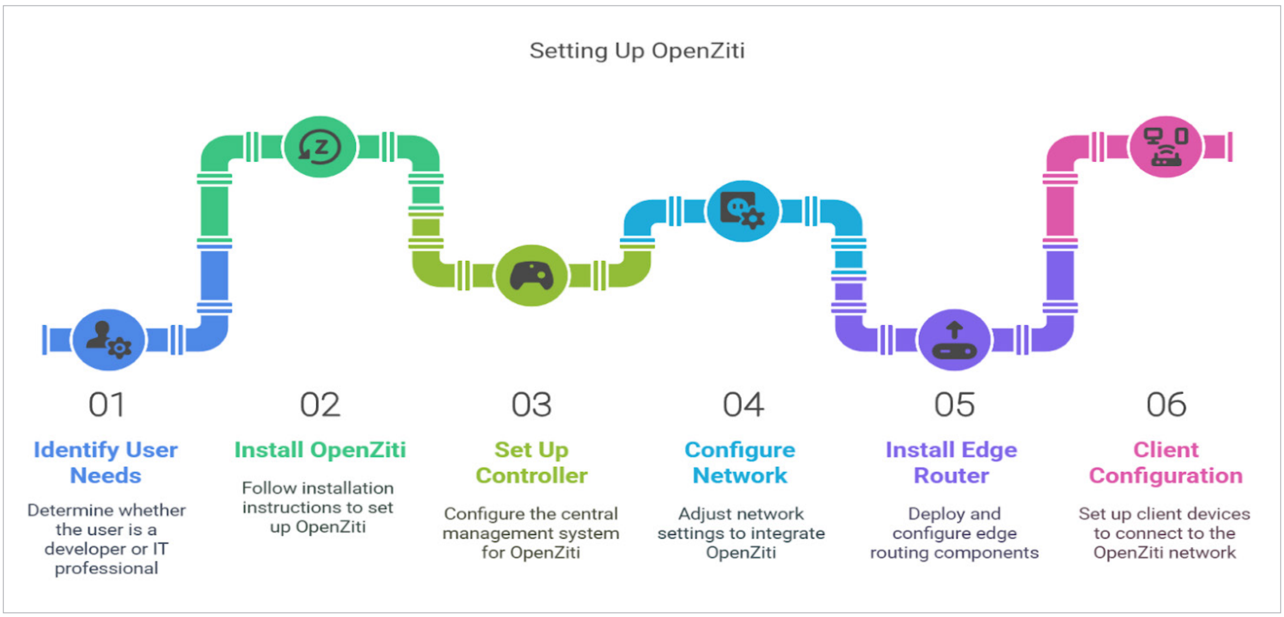

Figure 1: OpenZiti setup